

Quiz 1

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ENCN415-26S1 - Pavement Engineering

Status	Finished
Started	Friday, 27 March 2026, 7:05 PM
Completed	Friday, 27 March 2026, 9:00 PM
Duration	1 hour 54 mins
Marks	8.93/20.00
Grade	4.46 out of 10.00 (44.63%)

Question 1

Correct

Mark 0.50 out of 0.50

☐ [Flag question](#)

Question text

Soundness test is used to test aggregate resistance to weathering, wet and dry, freeze and thaw cycles and therefore it tests its durability.

Question 1

Select one:

True

False

Feedback

The correct answer is 'True'.

Question 2

Correct

Mark 0.50 out of 0.50

☐ Flag question

Question text

In the Sand Equivalent test, the higher the sand equivalent value; the less clean is the aggregates.

Question 2

Select one:

True

False

Feedback

The correct answer is 'False'.

Question 3

Correct

Mark 0.50 out of 0.50

☐ Flag question

Question text

Short term ageing represents

Question 3

Select one:

a.

Ageing during production only

b.

Ageing during production and construction

C.

Ageing over the design service life

d.

Ageing during construction only

Feedback

The correct answer is: Ageing during production and construction

Question 4

Incorrect

Mark 0.00 out of 1.00



Flag question

Question text

Aggregate blend consists of coarse aggregate, fine aggregate and mineral fillers with percentages 50, 45 and 5.0%, respectively. The bulk specific gravity of the coarse, fine and mineral fillers are 2.75, 2.85, 3.15, respectively. Determine the bulk specific gravity of the blend.

Answer:

Question 4

Feedback

The correct answer is: 2.81226

Question 5

Correct

Mark 0.50 out of 0.50

☐ Flag question

Question text

The higher the complex shear modulus of the asphalt binder, the stiffer is the binder.

Question 5

Select one:

True

False

Feedback

The correct answer is 'True'.

Question 6

Correct

Mark 0.50 out of 0.50

☐ Flag question

Question text

The lower the phase angle, the more viscous is the asphalt binder.

Question 6

Select one:

True

False

Feedback

The correct answer is 'False'.

Question 7

Partially correct

Mark 0.63 out of 5.00

☐ Remove flag

Question text

In a quality assurance process of one of the highway project; the road inspector cut a core and the following data was collected as shown below.

Dry mass of the core=1250 g

Saturated Surface Dry mass of the core= 1253 g

Submerged mass of the core =720 g

Bitumen content =6.0%

Effective bitumen content= 5.671%

Bulk Specific gravity of aggregate = 2.6680

Nominal maximum aggregate size is 12.5 mm

Design traffic is 12×10^6 ESA

Bitumen Specific Gravity = 1.035

Aggregate surface area is $6 \text{ m}^2/\text{kg}$

Based on the above data, determine:

1. Bulk Specific Gravity of the compacted mix
2. Answer 1 Question 7
3. use 4 decimal places
4. Percentage of bitumen absorption ($\%P_{ba}$)
5. Answer 2 Question 7
6. % use 2 decimal places
7. Maximum theoretical Specific gravity of the asphalt mix
8. Answer 3 Question 7
9. 2.55432.45632.78502.3500None of the answers
10. Percentage of voids in the total mix ($\%VTM$)
11. Answer 4 Question 7
12. % use two decimal places
13. Percentage of voids in the mineral aggregates ($\%VMA$)
14. Answer 5 Question 7
15. 17.3720.85None of the answers18.6419.79 %
16. Percentage of voids filled with bitumen ($\%VFB$)
17. Answer 6 Question 7
18. % use 2 decimal places
19. Percentage of the aggregate by volume of the total mix
20. Answer 7 Question 7

- 21. % use one decimal place
- 22. Field Performance of this mix is likely
- 23. Answer 8 Question 7
- 24. fatigue crackingflushing and ruttingNo specific issues are expectedRaveling, potholes and durability related issuesRutting

Feedback

Question 8

Correct

Mark 0.50 out of 0.50

☐ Flag question

Question text

The higher the creep stiffness, the less likely the binder will show thermal cracks.

Question 8

Select one:

True

False

Feedback

The correct answer is 'False'.

Question 9

Incorrect

Mark 0.00 out of 1.00

☐ [Flag question](#)

Question text

Asphalt mix with bulk specific gravity of 2.35, binder content 5.7%, aggregate bulk specific gravity 2.79 and bitumen specific gravity 1.03.

What is the percentage by volume of bitumen?

Answer:

Question 9

Feedback

The correct answer is: 13.005

Question 10

Correct

Mark 0.50 out of 0.50

☐ [Flag question](#)

Question text

In the BBR test, the higher the m-value, the more likely the binder will relax thermal stresses and therefore the better the binder performance in thermal cracking.

Question 10

Select one:

True

False

Feedback

The correct answer is 'True'.

Question 11

Partially correct

Mark 0.80 out of 4.00

Question text

The latitude at a location where a high-speed rural road is to be located at 30° . The seven-day average air temperature is 38°C and the low air temperature is -10°C . The standard deviation for the high temperature is 1.5°C and for the low temperature is 1°C . For a 97.5% reliability and design traffic of 20×10^6 ESA determine the following:

1. Calculate the average 7-Day maximum pavement temperature at 97.5% reliability,
2. Answer 1 Question 11
- 3.
4. Calculate the minimum pavement temperature at 97.5% reliability,
5. Answer 2 Question 11
- 6.
7. The suitable performance grade binder at 97.5% reliability for standard pavement section at normal speed
8. Answer 3 Question 11
9. PG58-16PG52-10PG64-10PG64-16PG46-34
10. The suitable performance grade binder at 97.5% reliability for slow traffic pavement section
11. Answer 4 Question 11
12. PG70-10PG46-34PG64-16PG58-16PG64-10
13. The suitable performance grade binder at 97.5% reliability for stopping traffic at weigh station pavement section
14. Answer 5 Question 11
15. PG76-16PG76-10PG70-10PG64-16PG70-16

Feedback

Question 12

Incorrect

Mark 0.00 out of 0.50



Flag question

Question text

LA abrasion test is used to measure aggregate toughness and the resistance to the impact of dynamic loads.

Question 12

Select one:

True

False

Feedback

The correct answer is 'True'.

Question 13

Correct

Mark 1.50 out of 1.50

☐ [Flag question](#)

Question text

A fine aggregate sample is characterized for its angularity; the volume of the measuring cylinder used in the test is 1000 cm^3 and the mass of the sand that filled the measuring cylinder is 1500g and its bulk specific gravity is 2.85. What is the fine aggregate angularity?

Answer:

Question 13

Feedback

The correct answer is: 47.37

Question 14

Partially correct

Mark 2.00 out of 3.00

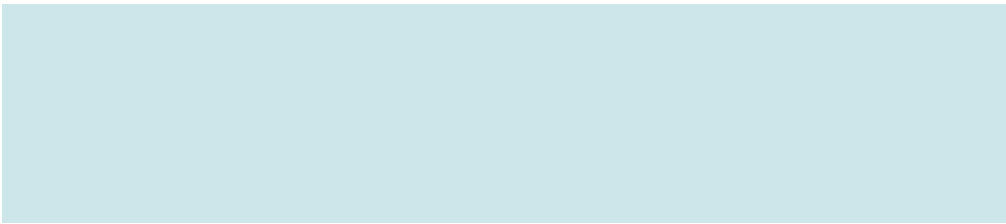
☐ [Flag question](#)

Question text

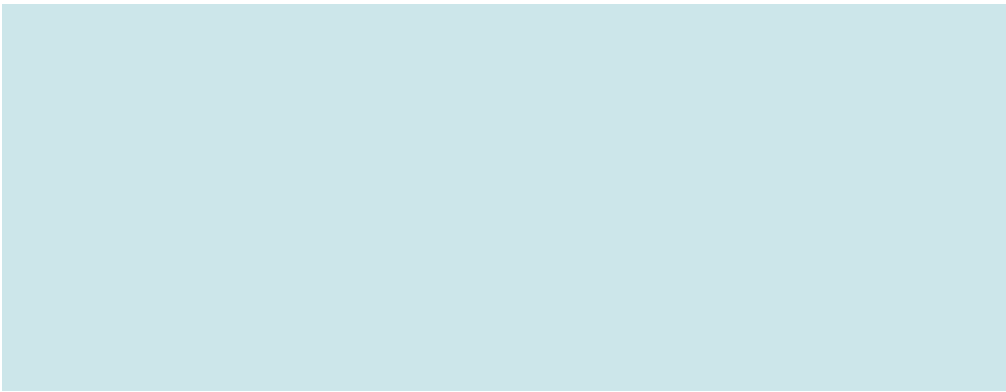
Two Binders A and B were tested for their rheological properties in the Dynamic Shear Rheometer (DSR) test. The DSR test was conducted at the average maximum 7-day pavement temperature and the complex shear modulus and phase angle were recorded as shown in Table 1. The DSR test was also conducted at the

intermediate pavement temperature and the complex shear modulus and phase angle were recorded as shown in Table 2.

Binder Type	Complex Shear Modulus (kPa)	Phase Angle
A	3.5	74.5
B	3.1	50.5



Binder Type	Complex Shear Modulus in kPa	Phase Angle
A	5000	45
B	6000	41



- 1. Rutting parameter for Binder A
- 2. Answer 1 Question 14
- 3. kPa Use two decimal places
- 4. Rutting parameter for Binder B

5. Answer 2 Question 14
6. kPa Use two decimal places
7. Which of the two binders is more permanent deformation resistant
8. Answer 3 Question 14
9. **ABNone of the answers**
10. Fatigue parameter for Binder A
11. Answer 4 Question 14
12. kPa Use two decimal places
13. Fatigue parameter for Binder B
14. Answer 5 Question 14
15. kPa Use two decimal places
16. Which of the two binders is more fatigue resistant
17. Answer 6 Question 14
18. **ANone of the answersB**

Feedback

Question 15

Correct

Mark 0.50 out of 0.50



Flag question

Question text

The higher the dissipation energy during loading cycles, the lower the fatigue and rutting distresses in the asphalt pavement.

Question 15

Select one:

True

False

Feedback

The correct answer is 'False'.

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Quiz 2

Quiz 2

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ENCN415-26S1 - Pavement Engineering

Status	Finished
Started	Saturday, 25 April 2026, 7:59 PM
Completed	Saturday, 25 April 2026, 10:00 PM
Duration	2 hours
Marks	19.50/36.00
Grade	5.42 out of 10.00 (54.17%)

Question 1

Partially correct

Mark 4.00 out of 11.00

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Question text

A Superpave mix design for a 15 million ESA is undertaken and the data shown below is recorded. Assume a 12.5mm nominal maximum aggregate size. The specific gravities of bitumen and aggregates are 1.05 and 2.785, respectively. Determine the following:

%P_b	5.5	6.0	6.5	7.0
G _{mb}	2.355	2.452	2.462	2.47
G _{mm}	2.588	2.589	2.583	2.566
% Dust (%P _d)	4.5	4.5	4.5	4.5
H _{ini}	129	131	131	128
H _{des}	112	113	116	115

1. Effective specific gravity at P_b=5.5%
2. Answer 1 Question 1
- 3.
4. Dust binder ratio at P_b=6.0%
5. Answer 2 Question 1
- 6.
7. Percentage of absorbed binder at P_b=6.5 %
8. Answer 3 Question 1
9. %
10. Effective binder content at (%P_{be}) at P_b=7.0%
11. Answer 4 Question 1
12. %
13. Percentage of voids in the mineral aggregates (%VMA) at pb=5.5%

14. Answer 5 Question 1

15. %

16. Percentage of voids filled with Bitumen (%VFB) at $P_b=6.0\%$

17. Answer 6 Question 1

18.

19. Percentage of voids in the compacted mix at $P_b=6.5\%$

20. Answer 7 Question 1

21.

22. Based on the %VTM and P_b shown below, the optimum binder content is

23. Answer 8 Question 1

24. 6.5 None of the above %5.5 %6.85 %7.0



What is the %VMA at optimum binder content

Answer 9 Question 1

%17.47 %18.5 %17.0 None of the above 20 %



What is the %VFB at the optimum binder content

Answer 10 Question 1

%80.0None of the above%77%75.565.5



Based on %VTM, VFB, %VMA, Does the mix design meet Superpave criteria

Answer 11 Question 1

yesNone of the answers%No%Not Clear

Feedback

Question 2

Correct

Mark 1.00 out of 1.00



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Question text

What happens when free lime (CaO) reacts with pozzolans (Al_2O_3 , SiO_2 , Fe_2O_3) in the presence of water?

Question 2

Answer

a.

It produces gases and causes swelling

b.

Neutralization of lime, reducing its effectiveness

c.

It forms calcium carbonate and increases plasticity

d.

Formation of CSH and CAH compounds, which are cementitious materials

Feedback

Your answer is correct.

The correct answer is:

Formation of CSH and CAH compounds, which are cementitious materials

Question 3

Partially correct

Mark 9.50 out of 19.00



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Question text

Based on the data shown in the table below for a Superpave design aggregate structure. Assume design traffic is less than 9 million ESA and the nominal maximum aggregate size is 19mm. Answer the following questions

Property	Blend 1	Blend 2	Blend 3
G_{mb}	2.451	2.465	2.467
G_{mm}	2.585	2.654	2.584
G_b	1.03	1.03	1.03
$\%P_b$	5.9	5.5	5.8
$\% \text{ Dust } (\%P_d)$	4.5	4.5	4.5
G_{sb}	2.657	2.667	2.7015
H_{ini}	127	135	124
H_{des}	113	114	118

For original un-adjusted blends properties

1. Percentage of air voids in the compacted mix for blend 1
2. Answer 1 Question 3
3. %

4. Percentage of air voids in the compacted mix for blend 2
5. Answer 2 Question 3
6. %
7. Percentage of air voids in the compacted mix for blend 3
8. Answer 3 Question 3
9. %
10. Percentage of effective binder content ($\%P_{be}$) for blend 1
11. Answer 4 Question 3
12. %
13. Percentage of effective binder content ($\%P_{be}$) for blend 2
14. Answer 5 Question 3
15. %
16. Percentage of effective binder content ($\%P_{be}$) for blend 3
17. Answer 6 Question 3
18. %
19. Percentage of voids in the mineral aggregates ($\%VMA$) for blend 1
20. Answer 7 Question 3
21. %
22. Percentage of voids in the mineral aggregates ($\%VMA$) for blend 2
23. Answer 8 Question 3
24. %
25. Percentage of voids in the mineral aggregates ($\%VMA$) for blend 3
26. Answer 9 Question 3
27. %
28. Percentage of voids filled with Bitumen ($\%VFB$) for blend 1
29. Answer 10 Question 3
30. %
31. Percentage of voids filled with Bitumen ($\%VFB$) for blend 2
32. Answer 11 Question 3
33. %
34. Percentage of voids filled with Bitumen ($\%VFB$) for blend 3
35. Answer 12 Question 3
36. %

For the adjusted air voids at 4.0% for the three blends 1,2 and 3 determine:

1. Percentage of binder content ($\%P_b$) for blend 1
2. Answer 13 Question 3
3. %
4. Percentage of binder content ($\%P_b$) for blend 2
5. Answer 14 Question 3
6. %
7. Percentage of binder content ($\%P_b$) for blend 3
8. Answer 15 Question 3
9. %
10. Percentage of effective binder content ($\%P_{be}$) for blend 1
11. Answer 16 Question 3
12. %
13. Percentage of effective binder content ($\%P_{be}$) for blend 2
14. Answer 17 Question 3
15. %
16. Percentage of effective binder content ($\%P_{be}$) for blend 3
17. Answer 18 Question 3
18. %
19. Percentage of voids in the mineral aggregates ($\%VMA$) for blend 1
20. Answer 19 Question 3
21. %
22. Percentage of voids in the mineral aggregates ($\%VMA$) for blend 2
23. Answer 20 Question 3
24. %
25. Percentage of voids in the mineral aggregates ($\%VMA$) for blend 3
26. Answer 21 Question 3
27. %
28. Percentage of voids filled with Bitumen ($\%VFB$) for blend 1
29. Answer 22 Question 3
30. %

31. Percentage of voids filled with Bitumen (%VFB) for blend 2
32. Answer 23 Question 3
33. %
34. Percentage of voids filled with Bitumen (%VFB) for blend 3
35. Answer 24 Question 3
36. %
37. the dust binder ratio for blends 1, 2 and 3, respectively are:
38. Answer 25 Question 3
39. 1.35, 1.45, 1.1 None of the answers 1.17, 1.25, 1.08 1.1, 1.25, 1.19 1.5, 1.3, 1.25
40. Based on Superpave criteria which bend of the three proposed blends best fit the requirements
41. Answer 26 Question 3
42. Blend 1 Blend 3 Blend 2 all of the three blends none of the three blends

Feedback

Question 4

Correct

Mark 1.00 out of 1.00

☐ Flag question

Question text

Which of the following are the three main techniques used in Hot In-Place (HIP) recycling of asphalt pavements?

Question 4

Answer

a.

None of the Answers

b.

Full-Depth Reclamation, Cold Recycling, Slurry Sealing

c.

Chip Seals, Cold Planing, Thin Overlay

d.

Surface Recycling, Repaving, Remixing

e.

Milling, Seal Coating, Crack Sealing

Feedback

Your answer is correct.

The correct answer is:

Surface Recycling, Repaving, Remixing

Question 5

Correct

Mark 1.00 out of 1.00



Flag question

Question text

Foam bitumen quality is measured by:

Question 5

Answer

a.

Half-life time only

b.

None of the answers

c.

Expansion ratio and half-life time

d.

Expansion ratio only

Feedback

Your answer is correct.

The correct answer is:

Expansion ratio and half-life time

Question 6

Correct

Mark 1.00 out of 1.00

☐ Flag question

Question text

Hot In-Place Recycling is a maintenance strategy that can effectively fix any type of pavement distress with high efficiency.

Question 6

Select one:

True

False

Feedback

The correct answer is 'False'.

Question 7

Correct

Mark 1.00 out of 1.00

☐ Flag question

Question text

What are the effects of adding lime to medium to highly plastic soils?

Question 7

Answer

a.

Increases plasticity, reduces workability, and causes swelling

b.

Has no significant effect on soil properties

c.

Decreases plasticity, improves workability, reduces swelling, and increases strength

d.

Only changes the colour and odour of the soil without altering its behaviour

e.

None of the Answers

Feedback

Your answer is correct.

The correct answer is:

Decreases plasticity, improves workability, reduces swelling, and increases strength

Question 8

Correct

Mark 1.00 out of 1.00

☐ Remove flag

Question text

The lower the voids in the uncompacted sand in the fine angularity test, the higher the aggregate angularity and vice versa.

Question 8

Select one:

True

False

Feedback

The correct answer is 'False'.

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Quiz #3_due on 15th May at 5:00

PM



Quiz #3_due on 15th May at 5:00 PM

Status	Finished
Started	Thursday, 14 May 2026, 3:05 PM
Completed	Friday, 15 May 2026, 4:49 PM
Duration	1 day 1 hour
Marks	24.00/40.00
Grade	6.00 out of 10.00 (60%)

Question 1

Incorrect

Mark 0.00 out of 10.00

☐ Flag question

Question text

In the three layer
pavement system
shown below, the
standard axle load of

80 kN and tyre pressure 750 kPa is acting on the pavement. Points b is located at the bottom of the asphalt layer exactly under the centre of the left outer tyre of the dual. Point c is located at the bottom of asphalt layer at the centre of the left dual tyres. Points d and e are exactly under points b and c respectively and at the top of the subgrade. All the three layers are assumed to be isotropic and the base layer to be sub-layered. Use Circly to analyse this pavement system. When you enter your answers, add negative sign for tensile strains only. For compressive strains, DON't add "+" sign.



1. Max tensile strain in microstrain units at point b at the bottom of asphalt layer
2. Answer 1 Question 1
3. use 2 decimal places
4. Max tensile strain in microstrain units at point c at the bottom of asphalt layer
5. Answer 2 Question 1

6. use 2 decimal places
7. Max tensile strain along x1-x1 axis at the bottom of the asphalt layer
8. Answer 3
Question 1
9. use 2 decimal places
10. Compressive strain in microstrain units at point d at the top of subgrade
11. Answer 4
Question 1
12. use 2 decimal places
13. Compressive strain in microstrain units at point e at the top of subgrade
14. Answer 5
Question 1
15. use 2 decimal places
16. Maximum Compressive strain in microstrain units at the top of the

subgrade
along X2-X2

17. Answer 6

Question 1

18. use 2 decimal
places

19. Max surface
deflection in
microns

20. Answer 7

Question 1

21. use 2 decimal
places

22. max
compressive
strain is
located at

23. Answer 8

Question 1

24. edge of outer
left tyre of the
left dualcentre
of the inner
tyre of the left
dualnone of
the
answerscentre
of the outer
left tyre of the
left dualcentre
of the left dual

Feedback

Question 2

Correct

Mark 10.00 out of 10.00

☐ Flag question

Question text

The Figure below, shows two unbound granular pavement sections A and B of different base thicknesses with a modulus at the top of the base of 500 MPa and a subgrade with a resilient modulus of 20 MPa, determine the value of the moduli for the sublayered base for these two sections:



For Section A

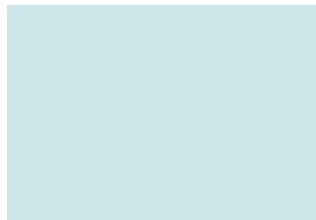
1. For Sublayer 1,
 E_1 in MPa
2. Answer 1
Question 2
3. Round up to
one decimal
place
4. For Sublayer 2,
 E_2 in MPa
5. Answer 2
Question 2
6. Round up to
one decimal
place
7. For Sublayer 3,
 E_3 in MPa
8. Answer 3
Question 2
9. Round up to
one decimal
place
10. For Sublayer 4,
 E_4 in MPa

11. Answer 4
Question 2
12. Round up to
one decimal
place
13. For Sublayer 5,
 E_5 in MPa
14. Answer 5
Question 2
15. Round up to
one decimal
place

For Section B

1. For Sublayer 1,
 E_1 in MPa
2. Answer 6
Question 2
3. Round up to
one decimal
place
4. For Sublayer 2,
 E_2 in MPa
5. Answer 7
Question 2
6. Round up to
one decimal
place
7. For Sublayer 3,
 E_3 in MPa
8. Answer 8
Question 2

9. Round up to one decimal place
10. For Sublayer 4, E_4 in MPa
11. Answer 9
Question 2
12. Round up to one decimal place
13. For Sublayer 5, E_5 in MPa
14. Answer 10
Question 2
15. Round up to one decimal place



Feedback

Question 3

Partially correct

Mark 8.00 out of 10.00

☐ Flag question

Question text

In the three layer pavement system shown below, the standard axle load of 80 kN and tyre pressure 750 kPa is acting on the pavement. Points b is located at the bottom of the asphalt layer exactly under the centre of the left outer tyre of the dual. Point c is located at the bottom of asphalt layer at the centre of the left dual tyres. Points d and e are exactly under points b and c respectively and at the top of the subgrade. Asphalt layer is assumed to be isotropic and the base and subgrade are assumed to be cross-anisotropic with the base layer to

be sub-layered. Use
Circly to analyse this
pavement system.
When you enter your
answers, add
negative sign for
tensile strains only.
For compressive
strains, DON't add "+"
sign.



1. Max tensile
strain in
microstrain
units at point
b at the
bottom of
asphalt layer

2. Answer 1
Question 3
3. use 2 decimal
places
4. Max tensile
strain in
microstrain
units at point c
at the bottom
of asphalt
layer
5. Answer 2
Question 3
6. use 2 decimal
places
7. Max tensile
strain along
x1-x1 axis at
the bottom of
the asphalt
layer
8. Answer 3
Question 3
9. use 2 decimal
places
10. Compressive
strain in
microstrain
units at point
d at the top of
subgrade
11. Answer 4
Question 3
12. use 2 decimal
places
13. Compressive
strain in
microstrain

units at point e
at the top of
subgrade

14. Answer 5

Question 3

15. use 2 decimal
places

16. Maximum

Compressive
strain in
microstrain
units at the
top of the
subgrade
along X2-X2

17. Answer 6

Question 3

18. use 2 decimal
places

19. Max surface

deflection in
microns as an
absolute value
(ignore the
sign)

20. Answer 7

Question 3

21. use 2 decimal
places

22. max

compressive
strain is
located at

23. Answer 8

Question 3

24. centre of the
inner tyre of
the left

larger of
the
answers centre
of the outer
left tyre of the
left dual edge
of outer left
tyre of the left
dual centre of
the left dual

25. The
assumption of
cross
anisotropic to
the
base course
and subgrade
under the
same axle
loads and
layer
thicknesses
and moduli
yield
compressive
strains at the
top of the
subgrade
compared to
the isotropic
assumption

26. Answer 9
Question 3

27. larger of
the
answers smaller
the same

28. The assumption of cross anisotropic to the basecourse and subgrade under the same axle loads and layer thicknesses and moduli yield tensile strains at the of the bottom of the asphalt layer compared to the isotropic assumption

29. Answer 10

Question 3

30. none of the answerslarger than the same smaller

Feedback

Question 4

Partially correct

Mark 6.00 out of 10.00

☐ [Flag question](#)

Question text



A rigid slab of
dimension 3.65 m by
12 m and depth of

0.25 m is subjected to wheel load of 40 kN and tyre pressure 800 kPa. If the modulus of subgrade reaction is 0.05 MPa/mm and the concrete slab has elastic modulus of 32 GPa and Poisson's ratio of 0.15, and the concrete coefficient of thermal expansion is 1.5×10^{-7} mm/mm/°C. The friction coefficient between the slab and subgrade is 0.45 and the steel yield strength is 300 MPa, The bulk specific gravity of concrete is 2.4, the maximum surface temperature of the concrete in the day time is 45 °C while the bottom temperature is 25 °C.

1. The radius of relative stiffness in mm
 2. Answer 1
- Question 4

3. use 2 decimal places
4. The corner stress due to the 40 kN acting on a tyre at the corner of the slab in kPa
5. Answer 2
Question 4
6. use 2 decimal places
7. The interior stress due to the 40 kN acting at the centre of the slab
8. Answer 3
Question 4
9. use 2 decimal places
10. The edge stress due to the 40 kN acting at the edge of the slab
11. Answer 4
Question 4
12. use 2 decimal places
13. The temperature correction factor C_x

14. Answer 5
Question 4
15. use 2 decimal
places
16. The
temperature
correction
factor C_y
17. Answer 6
Question 4
18. use 2 decimal
places
19. The warping
stresses (S_{xx})
at the centre
of the slab due
to thermal
temperature
gradient in kPa
20. Answer 7
Question 4
21. use 2 decimal
places
22. The warping
stresses (S_{yy})
at the centre
of the slab due
to thermal
temperature
gradient in kPa
23. Answer 8
Question 4
24. use 2 decimal
places
25. The steel in
 cm^2 per metre
width required

to keep cracks
intact.

26. Answer 9

Question 4

27. cm^2 use one
decimal place

28. Determine the
amount of
steel per
metre
required to tie
two adjacent
lanes.

29. Answer 10

Question 4

30. cm^2 use 2
decimal places

Feedback

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problems solution details or any required
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Quiz 4

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Quiz #4_Due 29 May at 5:00 PM



Quiz #4_Due 29 May at 5:00 PM

Status	Finished
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Started	Wednesday, 27 May 2026, 1:34 PM
Completed	Thursday, 28 May 2026, 2:37 PM
Duration	1 day 1 hour
Marks	19.29/20.00
Grade	9.64 out of 10.00 (96.43%)

Question 1

Correct

Mark 4.00 out of 4.00

Flag question

Question text



In the logging season, the traffic count recorded for the logging unit shown above is 10,000 passes in a year period. The average axle loads are shown above. If the annual traffic growth is 5%, the design period is 30 years and the traffic growth is geometric; determine:

1. Cumulative traffic growth factor over 30

years design
period

2. Answer 1

Question 1

3. use 2 decimal
places

4. The number of
ESA per one
passage of the
logging truck

5. Answer 2

Question 1

6. use 2 decimal
places

1. The
cumulative
design traffic
(DESA) for 30
year design
period

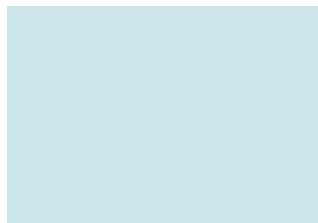
2. Answer 3

Question 1

3. use Scientific
format
(X.yzE0h)

If the traffic growth
with 5% will continue
for 20 years and it
will remain constant
($r\%=0$) for the remain
design period
determine:

1. Cumulative traffic growth factor over the 30 years design period
2. Answer 4 Question 1
3. use 2 decimal places
4. The cumulative design traffic (DESA) for 30 year design period
5. Answer 5 Question 1
6. use Scientific format (X.yzE0h)



Feedback

Question 2

Correct

Mark 5.00 out of 5.00

☐ Flag question

Question text

For a two-lane,
two-way highway, the
following traffic data
is as shown below:

AADT= 4000 vpd

Direction split 60/40

Percentage of heavy
vehicles 12%

Number of heavy
vehicle axle groups
per heavy vehicles is
2.5

Design period is 25
years

If the traffic grows geometrically at a growth rate 5.0% until the end of the design period.

Load (kN)	SAST	SADT	TAST	TADT	TRDT	QADT
45	0.21	0.08	0	0.1	0	0
55	0.13	0.08	0	0.1	0	0
65	0.15	0	0	0	0.05	0
75	0	0	0	0	0.1	0

Cumulative growth factor over the 25 year design period

Answer 1 Question 2

use two decimal places

Design traffic in terms of number of heavy vehicles axle groups

Answer 2 Question 2

use scientific format

X.xxE0X

For the traffic
distribution above,
determine ESA/HVAG

Answer 3 Question 2

use three decimal
places

The cumulative
design traffic in terms
of DESA

Answer 4 Question 2

use scientific format

X.xxE0X

Feedback

Question 3

Correct

Mark 6.00 out of 6.00

 [Flag question](#)

Question text

In the design of the flexible pavement cross section shown below. The design traffic distribution is as shown below and the design traffic is 12.5×10^6 HVAG.

Load (kN)	SAST	SADT	TAST	TADT	TRDT	QADT
45	0.21	0.08	0	0.1	0	0
55	0.13	0.08	0	0.1	0	0
65	0.15	0	0	0	0.05	0
75	0	0	0	0	0.1	0



Determine,

For the given traffic
distribution,
ESA/HVAG

Answer 1 Question 3

use three decimal
places

For 100 mm of
asphalt thickness, the
CDF for Asphalt

Answer 2 Question 3

use two decimal
places

For 100 mm of
asphalt thickness, the
CDF for Subgrade

Answer 3 Question 3

use three decimal
places

The design thickness
of asphalt to support
the design traffic

Answer 4 Question 3

mm round up to a
whole number

Feedback

Question 4

Partially correct

Mark 4.29 out of 5.00

☐ Flag question

Question text

In the falling weight deflectometer, if 80 kN weight was dropped on a three layers pavement as shown below. The deflection bowl was measured by using seven geophones and the data was recorded as shown in the table below. Calculate the pavement layers moduli using trial and error (Assume all layers isotropic and no sublayering).

Sensor Offsets (mm)	0	20	400	60	800	100	120			
		0		0		0	0	1400	1600	1800

Deflection
(microns)

64	57	500.	43	384.	339	299.			
2	4	4	8	7		7	265.	236.	211.
							9	9	9



Determine:

1. Deflection ratio,
2. Answer 1
Question 4
- 3.
4. Surface Curvature

5. Answer 2
Question 4
6. 45None of the
answers104.85
0606880
7. Surface
curvature
Index (SCI)
8. Answer 3
Question 4
- 9.
10. From First
Principles,
estimate
subgrade
modulus
11. Answer 4
Question 4
12.
53.2452.65743
570

Use BAKFAA
Determine

1. Asphalt Layer
Modulus(E_1) in
MPa,
2. Answer 5
Question 4

3.

474040003000

35002.6574
4. Base Layer
Modulus(E_2) in
MPa,
5. Answer 6
Question 4
- 6.
7. Subgrade
Layer
Modulus(E_3) in
MPa,
8. Answer 7
Question 4
- 9.

Feedback

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 problems solution details or any required

attachment Midterm
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